University of Eloued

Series 07
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Destination: Pro.Atallah amore and Dr. Ghanaim Zouhira

Random Forest approach

Party one:
Using a small dataset

The Random Forest algorithm is an advanced ensemble learning technique that mitigates the overfitting risks inherent in single decision trees. By leveraging Bootstrap Aggregating (Bagging), the algorithm trains a diverse "committee" of shallow trees on random subsets of the data, deliberately excluding approximately 36.8% of observations to serve as an internal validation set. Furthermore, by restricting node splits to a randomized subset of variables (Feature Randomness,$m \approx \sqrt{P}$, the model prevents dominant variables from dictating every tree's architecture. The final classification is determined by a majority vote, yielding a highly stable, generalized model ideal for complex decision-making tasks such as financial risk assessment.

Applied Python Coding Exercise

Dataset Specifications The following dataset contains 6 client observations used to predict credit risk.

Observation	(X1) Number of Late Payments	(X2) Credit Risk Level	Class (Risk Category)
A	1	1	-1 (Low)
B	2	1	-1 (Low)
C	1	2	-1 (Low)
D	4	4	+1 (High)
E	5	4	+1 (High)
F	4	5	+1 (High)

1. DataFrame Initialization: Write a Python script using the `pandas` library to construct the dataset above into a structured DataFrame. Separate the explanatory variables (X_1, X_2) into a feature matrix `X` and the target variable into a vector `y`.

2**. Model Configuration**: Import the appropriate ensemble module from `scikit-learn`. Instantiate a Random Forest Classifier configured to replicate the theoretical manual exercise: set the number of trees to 3, restrict the maximum features evaluated at each split to 1, enable Out-of-Bag (OOB) scoring, and set a fixed random state for reproducibility.

3. Training and Internal Validation: Fit the instantiated model to your feature matrix `X` and target vector `y`. Write the code required to extract the model's internal `oob_score_`, calculate the final OOB Error Rate, and print the result.

4. Majority Vote Prediction: A new client profile is submitted to the system with 4 late payments (X_1 = 4) and a credit risk level of 2 (X_2 = 2). Write the code to format this new data point correctly and use the trained forest to predict the client's final risk class.
5. Tree Extraction (Optional Advanced Step): To audit the ensemble's internal architecture, write a brief loop using the model's `estimators_` attribute to print the maximum depth and the number of nodes for each individual tree generated within the forest.

Solution:
You can find the full solution in the below Link:
Git_Hub:

https://github.com/atman500/atman500/blob/main/Random_forest_using_simple_example.ipynb

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Second Step: using a large dataset:

Name of the file; Bank.csv

Solution:
You can find the full solution in the below Link:
Git_Hub:

https://github.com/atman500/atman500/blob/main/Random_forest_using_Big_dataset.ipynb